

AIRPORT PROPERTY MANAGEMENT SYSTEM

SRS Document

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1.Introduction

1.1. Purpose

This Software Requirements Specification document details the functional, non functional and structural requirements for the Airport Property Management System. It provides a complete design for development and evaluation.

1.2. Scope

The APMS is a web based software application designed to manage commercial shops, physical properties and rental contracts inside an airport terminal. The system automates administrative workflow, simplifies rental agreement calculations, secures data logs and tracks real time terminal point of sale revenues to eliminate the inefficiencies of manual paperwork and spreadsheet management.

2. Overall Description

2.1. Product Perspective

The system follows a 3-tier decoupling architecture consisting of a React-based client user interface running on <http://localhost:5173>, a Node.js/Express.js application backend server container running on <http://localhost:5000> and a secure local MySQL relational database execution model named dbdutyfreepos.

2.2. Product Functions

The system is divided into 7 distinct administrative and operational views:

- User Login Pages : Secure access gateway for system administrators.
- Live Sales Transaction Terminal : Real time monitoring ledger for terminal wide retail transactions.
- Company Profile Management : Record keeping and setup for registered parent corporations.
- Shop Profile Module : Configuration tool linking retail sub-shops to their corporate parents.
- Rental Agreement Management : Legal tenancy contract setup, deposit ledger and rate configuration panel.
- Property Profile Inventory : Spatial mapper and real estate slot tracking tool.
- Analytical Reporting Engine : Timed database filtering tool for business audits and metrics.

3. System Features & Functional Requirements

3.1 User Login Module

- The system shall display input text boxes for USERNAME and PASSWORD inputs.
- The backend must authenticate user entries dynamically against the tbllogin data table.
- Upon successful verification, the system shall issue a secure JSON Web Token valid for 1 hour to protect application routes.

3.2 Live Sales Transaction Module

- The system shall display a live tabular database log showcasing the invoice numbers, transaction shop names, corporate company names, locations and unique DayIDs
- The interface shall provide relational dropdown filters allowing operators to sort the ledger data grids by a specific Corporate Company or Tenant shop
- The page shall provide a REFRESH LOGS trigger button to re-fetch and update current sales listings from the database server.
- The user shall be able to click a View actions button to inspect itemized transaction data points.

3.3 Company Management Module

- The interface shall lock the Company ID field to be Auto Generated by DB.

- The operator shall be able to register new parent entities by inputting the Company Name, Corporate Email, Headquarters Address, and a Contact Hotline.
- The system shall display a Registered Corporate Ledger highlighting company rows with an active status tag pill (e.g., Active).
- The table must provide action icons allowing users to click a Pencil icon to modify data rows or a Red Trash icon to clear a corporation profile.

3.4 Shop Management Module

- The user shall be able to configure sub-shops by picking an established parent entity using a Main Company dropdown selection field.
- The system shall capture specific operational parameters including the Shop Name, Email Address, Contact No, Doc Count, and data sync modes (Integration types like POS / Manual / API).

3.5 Rental Agreement Engine

- The contract setup layer must enable relational linkage by binding a designated Main Company and Shop Name to a specific Property Name and Location ID.
- The operator shall explicitly define lease limits using Agreement Starts on and Agreement Ends On date calendars.
- The engine shall log baseline tenancy financial rules including the Minimum STO Level (in LKR), Monthly Amount (Fixed Rent), and the Confee Rate % (Concession Fee commission percentage calculated on shop sales streams).

- The interface must capture deposit securities via Payment Method, Deposit Type, Bank Deposit values, and a dedicated Expiry Date calendar.

3.6 Property Inventory Module

- The operator shall catalog property slots under Spatial Metrics parameters including Property Name/No, Area (Sqft) sizing, and unique Location ID codes.
- The interface shall allow administrators to select a structural state using a Leased Status selection box (e.g., No (Available) or Yes (Leased)), assign a specific Chart Color hex code, and enter future development details inside a Proposed Property text field.
- The Airport Property Registry table must support inline row modifications, allowing administrators to click a Pencil Edit button to switch fields into editable text/dropdown inputs, then click Save to commit updates or Cancel to abort changes.

3.7 Analytical Reporting Engine

- The interface shall render four analytical command blocks: Daily, Monthly, Shopwise Monthly, and Sales & Invoice Count.
- The dashboard must execute automated relational reporting queries when an administrator picks temporal calendar dates and clicks evaluation triggers like VIEW DAILY, VIEW MONTHLY, or VIEW SHOPWISE MONTHLY.
- The system must calculate and output precise metric data grids displaying minimum limits, parent company tags, shop name linkages, and daily sale amounts mapped to the LKR currency.

- The screen shall provide instant export options allowing users to download generated summary metrics by clicking a red PDF icon button or a green EXCEL spreadsheet document button.

4. External Interface Requirements

4.1 User Interfaces

- The graphical responsive frontend panels must render dark-theme dashboards, interactive form grids, drop-down option lists, and explicit data validation alerts to guide data handling workflows.

4.2 Database Interface Specifications

- The persistence layer shall be managed by a MySQL instance. Relational table requests will target schemas like `tbllogin`, `tblcompanies`, and transactional sales tables structured cleanly under the database named `dbdutyfreepos`.

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Security & Access Control

- All operational route controllers must be structurally protected behind token validation mechanisms checking key environments (`process.env.JWT_SECRET`) to intercept unauthorized external API requests.

5.2 Performance & Reliability

- The system backend architecture (`server.js`) must continuously run on Port 5000 and ensure low query execution latency during intensive real-time reporting generation and high-concurrency data requests.

